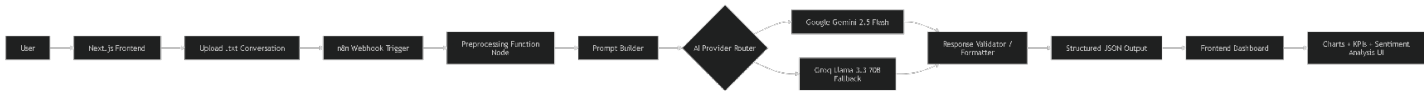


Conversation Sentiment Analyzer — Submission Summary

A full-stack AI application that takes a call transcript, analyzes it through an orchestrated AI pipeline, and presents sentiment, emotion, and call-center KPIs on an interactive dashboard.

Architecture



UI → n8n → AI. The frontend never calls an AI provider directly. Every analysis request goes through an n8n orchestration layer, which owns the prompt, the provider choice, and the response shaping. This keeps the AI logic swappable — changing models, providers, or adding new analysis steps requires no frontend changes or redeployment.

Resilience: if the primary provider (Gemini) fails for any reason — quota exhaustion, timeout, outage — the workflow automatically detects the failure and retries the identical prompt against a second provider (Groq, running Llama 3.3 70B), with no user-visible disruption. This was built and verified by deliberately breaking the primary provider's credentials and confirming the pipeline still returned a complete, correctly-shaped result.

Stack

Layer	Technology
Frontend	Next.js (App Router), TypeScript, Tailwind CSS, Recharts
Auth	Session-based via signed JWT cookies, bcrypt-hashed credentials (no database)
Orchestration	n8n (webhook → prompt construction → AI call → validation → response)
Primary AI	Google Gemini 2.5 Flash
Fallback AI	Groq (Llama 3.3 70B)
Hosting	Vercel (frontend), n8n Cloud (orchestration)

Meeting the Output Expectations

- Overall sentiment** (Positive / Negative / Neutral) with a confidence score and a written explanation grounded in the actual transcript content.
- Sentence-level sentiment** for every line of the conversation, each tagged with speaker, sentiment, emotion, intensity, and a short reasoning phrase naming the specific cue that drove the call.
- Call-center KPIs** chosen for relevance to phone support specifically rather than generic text sentiment: CSAT estimate, resolution status, agent empathy score, sentiment trend

across the call, escalation risk (with reasoning), key issues raised, suggested follow-ups, and talk-time balance between speakers.

How AI Quality Is Addressed

Two deliberate prompt-design choices target the “logical accuracy” and “clear reasoning” evaluation criteria directly, rather than treating them as implicit:

- **Reasoning is a required output field, not an afterthought.** The model is required to justify the overall sentiment, the escalation risk assessment, and every individual sentence's sentiment call. A label with no justification is not a valid output under this schema.
- **The prompt explicitly forbids the most common LLM sentiment failure mode** — defaulting to a safe Neutral/Positive label under ambiguity. It instructs the model to be analytically honest and to ground reasoning in specific transcript content rather than generic statements.

This makes the model's reasoning inspectable at every level of the output, so accuracy can be judged by reading why a call was made, not just trusting the label.

Creativity

- **Sentiment waveform** — sentence-by-sentence sentiment plotted as a continuous signal across the call, styled like an audio/ECG waveform, with hover tooltips showing the sentence and its reasoning.
- **Emotion distribution chart** — counts across nine emotion categories (joy, frustration, anger, sadness, satisfaction, confusion, anxiety, gratitude, neutral), not just three-way sentiment.
- AI-generated conversation summary.
- **Six-plus KPIs** beyond the minimum ask, including escalation risk with reasoning, talk-time balance, and suggested action items.
- **Dual-provider resilience**, an architecture-level creativity choice that also demonstrates production-mindedness.

Known Limitations

- **No database** — by design, to match the assignment's “basic auth is fine” scope. A single operator account is configured via environment variables, with the password stored as a bcrypt hash rather than plaintext. Swapping in a real users table later requires changing one function, not restructuring the app.
- **n8n Cloud's free trial** occasionally deactivates idle workflows; this is a trial-tier behavior, not an application defect, and does not occur on a paid plan or a self-hosted instance.

Live Demo

- **Frontend:** <https://sentiment-analyzer-ten-beta.vercel.app/login>

Username: "admin"
password: "123456789"

- **Github:** <https://github.com/Hardik-exe/Sentiment-analyzer>